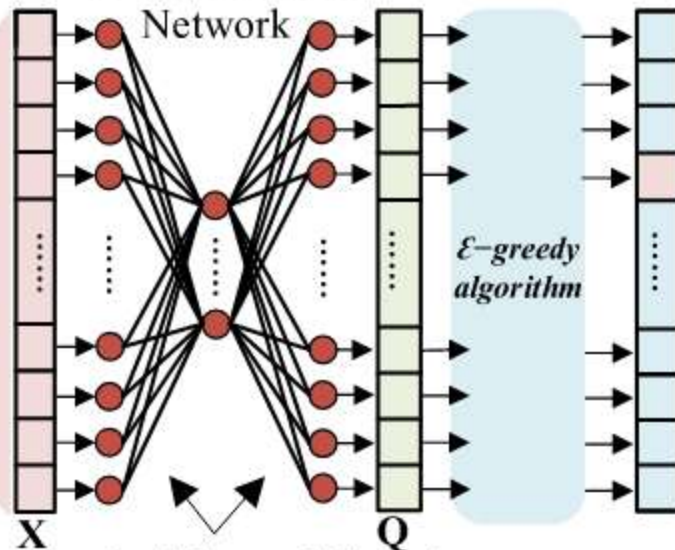


Components definition

	L_+	L_-		V_{i+}	V_{i-}
L_+	0	1	0	0	0
L_-	0	0	0	0	0
	0	0	0	0	0
	0	0	0	1	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
V_{i+}	0	0	0	0	0
V_{i-}	0	0	0	0	0

M_{adj}

Q-value Neural Network



	L_+	L_-		V_{i+}	V_{i-}
L_+	0	1	0	0	0
L_-	0	0	0	0	0
	0	0	0	0	0
	0	0	0	1	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
V_{i+}	1	0	0	0	0
V_{i-}	0	0	0	0	0

Prior design constraints

NN training $\phi \leftarrow \phi - \alpha \frac{\partial \text{Loss}}{\partial \phi}$

Current Loop Extraction

Voltage values for volt-second balancing examination



Reward

R_x

Design specifications

$$\text{Loss} = (q_k^* - q_k)^2$$

Loss function

$$q_k^* = R_x + \gamma \cdot \max_{\phi} f_{\phi}(\mathbf{X}')$$

Q-learning RL Algorithm